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# PALEOMAGNETISM OF TRIASSIC PARSORA SANDSTONES FROM INDIA*

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The paleomagnetism of fifty samples of red sandstones collected from three different sites belonging to the Parsora formation of Upper Triassic age has been studied. The natural remanent magnetization of the samples showed a wide scatter. The intensity of magnetization of the samples ranged from $3\times10^{-6}$ to $50\times10^{-6}$ gauss. The stability of magnetization was established using alternating field demagnetization up to 600 oerstedt peak fields as well as thermal demagnetization in zero field up to 250 °C. The thermal demagnetizing technique was found to be more suitable as compared to a.c. demagnetization. The mean declination and inclination for three sites after thermal demagnetization was 320°E and −39° (up). The results are consistent with India being in the southern hemisphere during Upper Triassic times and indicate that practically no drift took place between Upper Triassic and Upper Permian times.

## 1. Introduction

Paleomagnetic studies made so far on mesozoic and cenozoic formations in India have given a strong support to the hypothesis of continental drift for the Indian landmass since Upper Permian times. The formations which have been studied particularly in detail are the Deccan traps of Cretaceous to Eocene age (CLEGG *et al.* (1956), DEUTSCH *et al.* (1959), VERMA *et al.* (1963) and SAHSRABUDHE (1963), the Tirupati sandstones of Lower to Middle Cretaceous age (VERMA and PULLAIAH (1967), the Rajmahal traps of Lower to Middle Jurassic age (CLEGG et al. (1958), the Sylhet traps probably of Jurassic age (ATHAVALE *et al.* (1963), RADHAKRISHNAMURTY (1964)) and Kamthi sandstones of Upper Permian age (VERMA and BHALLA (1968)). There has been a need for a thorough study of the Gondwana formations in India to cover the important period of Cenozoic and Mesozoic during which the major part of the continental drift appears to have taken place. This period is also vitally important for interpreting the results of continental drift of the Gondwana continents. ATHAVALE *et al.* (1968) have recently reviewed the paleomagnetic data from India for all formations including the Precambrian.

Detailed paleomagnetic studies of Gondwana formations are important from the point of view of understanding the mechanism of continental drift, the rate of drift through the geologic times, the actual period of the start of the drift, the nature of the geomagnetic field and the paleosecular variation throughout this period.

Some of the formations in Gondwana basins of India ranging in age from Cretaceous to Permian are quite suitable for paleomagnetic studies on account of their adequate intensity. Recent studies made on paleomagnetism of the Parsora formation of Upper Triassic age from the Johilla coal field situated in the southwestern part of the Rewah Gondwana basin have given interesting results regarding the continental drift of the Indian landmass. The results are discussed in this paper.

## 2. Geology of the area

The Johilla coal field forms a part of the south Rewah Gondwana basin which runs in a general E–W direction from near Ramanuj Gunj (23°45′N, 83°45′E) in the East to near Chandia (23°40′N, 80°45′E) in the West. The basin is nearly 50 km wide. It is filled with sediments ranging in age from Carboniferous to Cretaceous. The stratigraphic sequence of rocks in the basin is given in table 1. The Parsora formation is characterised by grey to buff coloured sandstones within which red coloured sandstones of 6 to 8 ft.

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TABLE 1
Stratigraphic sequence in the Johilla coal field according to KRISHNAN (1960)

| | |
|---|---|
| Bansa Beds | Upper Jurassic |
| Athagarh Sandstones | Lower Jurassic |
| Tiki, Chicharia, Parsora | Triassic |
| Himgir | Upper Permian |
| Barakar | Lower Permian |
| Talchir | Carboniferous |

thickness occur. Regarding the age of the formation, the following evidence has been presented.

COTTER (1918), while classifying the Gondwana formations of India, has identified several fossils such as Danaeopsis Hughesi, Dicrodium Odontopteroides, Asplenium Whitbynse and Cordaites Hislopi near the village of Parsora (our site 3) and assigned the age of Rhaetic to this formation. SEWARD (1933) while examining some fossils exposed about 6 miles north of Parsora (23°26'N, 81°06'E) has stated that "in the highest degree it is probable that the Parsora flora is not younger than Lower Triassic and the most abundant species lead me to believe that it agrees more closely with late Permian flora of the northern hemisphere than with any Mesozoic floras". Detailed study of fossils from Parsora formation particularly near our sampling sites has been made by SAHNI and RAO (1958). They have identified such fossils as Schizoneura Gondwananensis, Glossopteres Indica, G. Communis, G. Retifera, G. Ampla and others from Johilla river (near our site 2), Dicordium Thinfeldia Hughesi from Ghorari Nala (near our sites 1 and 3) as well as from

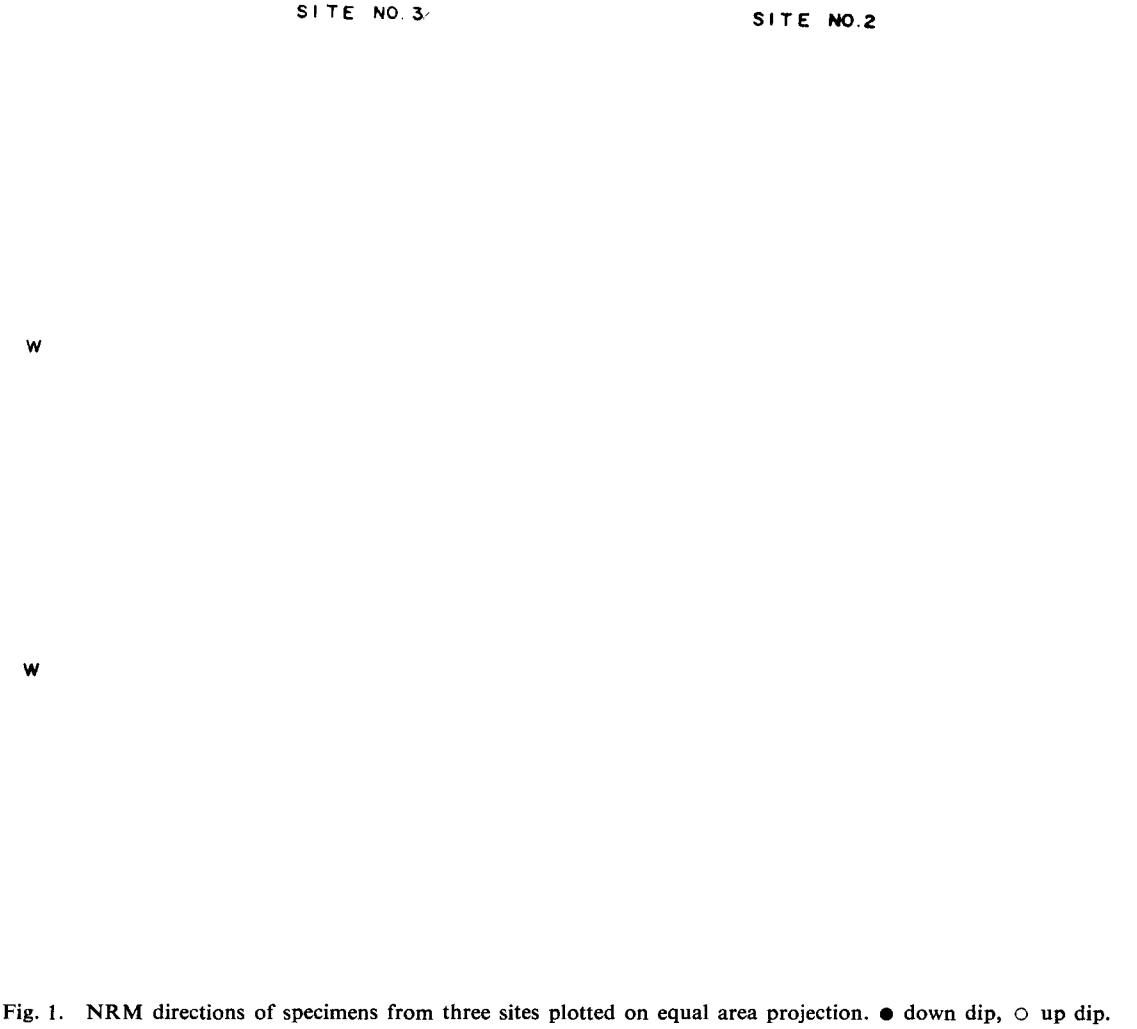



Fig. 1. NRM directions of specimens from three sites plotted on equal area projection. ● down dip, ○ up dip.

other near by localities. These authors have examined all the detailed fossil evidence of their own as well as previous workers and have concluded that the Parsora formation is of Rhaetic to Keuper (Upper Triassic) in age.

### 3. Sampling

According to Fox (1931) the stratigraphic thickness of the Parsora formation between the village Parsora (our site 3) and Birsinghpur Pali (our sites 1 and 2) is about 800 ft. The paleomagnetic results discussed here are from samples collected from three different sites which are located within a few kilometers of the town of Birsinghpur Pali (23°26′N, 81°02′E). Fresh samples were collected as far as possible either in river beds or in small hillocks. The main river which cuts through the Parsora formation is the Johilla river which has a prominent tributory by the name of Ghorari Nala.

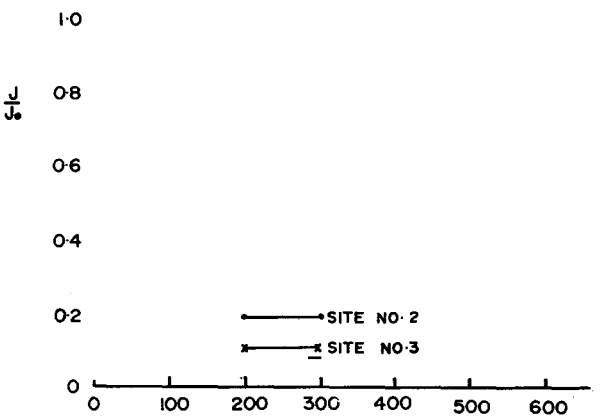



Fig. 2. Normalised intensity of some representative specimens plotted against peak alternating fields for sites 1, 2 and 3.

At site no. 1 twenty hand samples were collected from a small hill near the village Dhorai about one mile north of Birsinghpur Pali. The strike of the formations here was N 60°E and the dip towards

Fig. 3. Change in the direction of magnetization as a result of alternating field demagnetization for some representative specimens from each site.

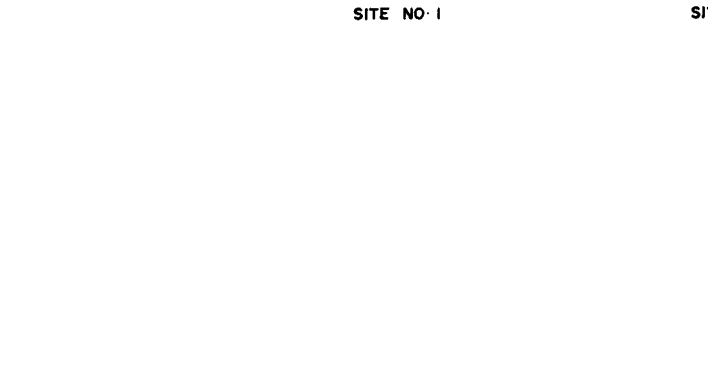



Fig. 4. Mean directions of the samples from sites 1 and 3 after a.f. demagnetization in peak fields of 400 oerstedt.

TABLE 2

Paleomagnetic parameters for the Parsora sandstones of Triassic age

| Locality | Site no. | $N$ | $E$ | $n$ | Mean direction after a.f. cleaning | | $K$ | Mean direction after thermal cleaning | | $K$ |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | $D$(°E) | $I$(°) | | $D$(°E) | $I$(°) | |
| Dhaurai 23°23′N 81°02′E | 1 | 20 | — | 68 | 328 | −35 | 90 | 319 | −39 | 190 |
| Johilla river 23°26′N 81°01′E | 2 | 18 | 1 | 46 | — | — | — | 317 | −42 | 94 |
| Ghorari Nala 23°26′N 81°06′E | 3 | 11 | — | 27 | 323 | −34 | 12 | 322 | −34 | 33 |
| Mean | | 3 | — | — | — | — | — | 320 | −39 | 200 |

$N$ number of samples included in the calculation of the mean direction.
$E$ number of samples excluded in the calculation of the mean direction.
$n$ number of specimens (discs) studied.
$D$ declination, east of true north (degrees).
$I$ inclination, north seeking pole up (-ve).
$K = (N-1)/(N-R)$, estimate of Fisher's precision parameter.

| Locality | Site no. | $R$ | $\Delta$ | $a_{95}$ | VGP (North pole) | | $d_p$ (°) | $d_m$ (°) | $\lambda$ (°S) |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | $\lambda_p$ (°N) | $L_p$ (°W) | | | |
| Dhaurai 23°23′N 81°02′E | 1 | 19.9 | — | 2.3 | 29 | 54 | 1.4 | 2.4 | 22 |
| Johilla river 23°26′N 81°01′E | 2 | 17.82 | — | 3.4 | 27 | 55 | 2.5 | 2.4 | 24 |
| Ghorari Nala 23°26′N 81°06′E | 3 | 10.7 | — | 7.3 | 33 | 54 | 4.7 | 8.5 | 19 |
| Mean | | — | 9° | 5.7 | 30 | 55 | 4.0 | 6.8 | 22 |

$R$ length of the resultant of $N$ unit vectors.
$a_{95}$ radius of the circle of confidence (95%).
$\Delta$ angular dispersion factor.
$\lambda_p$ latitude of the virtual geomagnetic pole.
$L_p$ longitude of the virtual geomagnetic pole.
$\lambda$ paleolatitude of the site.
$d_p$, $d_m$ semi axis of the ovals of confidence about the paleomagnetic pole position at 95% probability level.

S 30°E. At site no. 2 seven hand samples were collected and twelve cores (2″ to 3″ long) and were drilled using a portable core drilling equipment similar to the one discussed by DOELL (1967). Here the strike and the dip of the formation were N 60°E and 10° (towards S 30°E) respectively. At site no. 3 six hand samples and 5 cores were obtained. Here the strike was N 30°E and dip 10° towards S 60°E respectively. At each site the stratigraphic thickness sampled was about 2 m. The exposures at site 2 and 3 are particularly fresh having been cut and exposed recently by the rivers.

## 4. NRM measurements

Three to four discs of 2.5 cm diameter were studied from each hand sample and 1 to 2 discs from the cores. The NRM directions on the specimens (discs) are plotted in polar equal area projection for sites 1, 2 and 3 in fig. 1. The directions from site 2 show some streaking effect, towards the present dipole field similar to that observed by CREER (1959).

The NRM directions for most of the samples were consistent between themselves except for a few cases. Six samples from site 1 and five from site 2 showed a large degree of scatter within themselves. The mean intensities of NRM at the three sites were $5\times10^{-6}$ gauss (site 1), $23\times10^{-6}$ gauss (site 2) and $9.5\times10^{-6}$ gauss (site 3). The total range of the intensities was $3.1\times10^{-6}$ to $50\times10^{-6}$ gauss. The mean values of the susceptibility for site 1, 2 and 3 were $9.84\times10^{-6}$, $18.7\times10^{-6}$, $11.2\times10^{-6}$ gauss/oerstedt respectively.

## 5. Stability tests

### 5.1. *Demagnetization in alternating fields*

The scatter in the NRM directions could be due to several causes, particularly viscous components of magnetization acquired by the samples in the present Earth's field, or due to chemical changes produced in the magnetic minerals after their deposition which resulted in a change of their original magnetic directions.

In order to remove the effects of viscous magnetization acquired by the samples in the present Earth's field some pilot specimens from each site were first cleaned in alternating fields of peak values 50, 100, 200, 400 and 600 oerstedt. The response of the specimens to the alternating field (a.f.) demagnetization was different in the case of each site. For site 1 and 3 this technique was useful in reducing the scatter in peak fields upto 600 oerstedt and 200 oerstedt respectively, indicating differences in coercivity of the specimens for these sites. For site 2 there was very little change up to 400 oerstedt fields. The specimens from site 1 showed the least change in intensity after demagnetization up to 600 oerstedt peak fields. Fig. 2 shows typical a.f. demagnetization curves of normalized intensity at various field strengths. A typical example of the change in the magnetic directions as a result of this process is shown in fig. 3 for all three sites. All specimens from sites 1 and 3 were cleaned in peak fields of 400 oerstedt and the magnetic directions for each sample after cleaning are shown in fig. 4. The precision parameter using FISHER's (1953) statistics were 90 and 12 for sites 1 and 3 respectively. Mean remanent directions for sites 1 and 3 after cleaning are given in table 2.

### 5.2. *Thermal demagnetization*

The stability of the magnetization of the sandstones was further tested by heating the samples in a sedimentary furnace enclosed in a Helmholtz coil system designed after SCOTT (1957). The uniform field space (with a uniformity of about 0.1% of the field at the

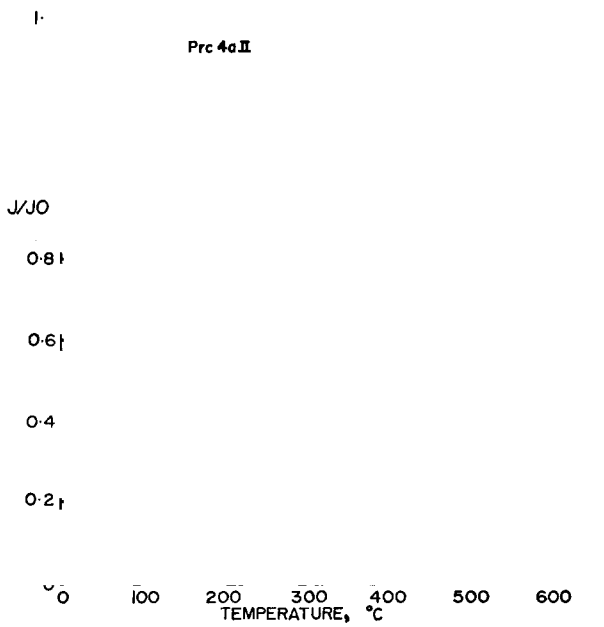



Fig. 5. Thermal demagnetization curves of normalized intensity for some specimens up to 600 °C.

N

Fig. 6. Changes in the direction of magnetization for some specimens from each site as a result of thermal demagnetization at various temperatures in zero field.

centre) was about 20 cm in diameter. Thermal demagnetization for some specimens from each site was done in steps of 125, 175, 250, 350, 450 and 550 °C. Typical thermal demagnetization curves for some specimens are shown in fig. 5. A change in the magnetic directions as a result of thermal cleaning at different temperatures for three sites is shown in fig. 6. Several heating runs for different specimens from the three sites showed that heating up to about 175 °C temperature was sufficient to obtain stable directions. The magnetic directions obtained after heating up to this temperature were taken to be representative for the original directions and were used for the interpretation of paleomagnetic results. Thermally cleaned directions for samples from the three sites are plotted on a polar equal area projection in fig. 7 and the mean values for each site are given in table 2.

## 6. Results and discussion

The mean directions from the various sites after alternating field and thermal cleaning together with other paleomagnetic parameters are given in table 2. The directions were corrected for the dip of the formations. FISHER's (1953) statistics have been used in the representation and analysis of the data. In the calculation of the mean directions of the sites, unit weight was given to each sample and to each core drilled at the site. It is noticed that the precision parameter $K$ increased from 90 after a.c. cleaning to 190 after thermal cleaning for site 1. The corresponding change for site 3 is from 12 to 33. For site 2 the change of the precision parameter is quite appreciable. Overall the thermal cleaning technique proved to be more useful in removing the effects of secondary magnetization although effective cleaning by a.c. also seems possible in certain cases. The thermal demagnetization technique for magnetic cleaning of sedimentary rocks has been used successfully by IRVING et al. (1961, 1965), IRVING (1963), CHAMALAUN et al. (1963), OPDYKE (1964) and others.

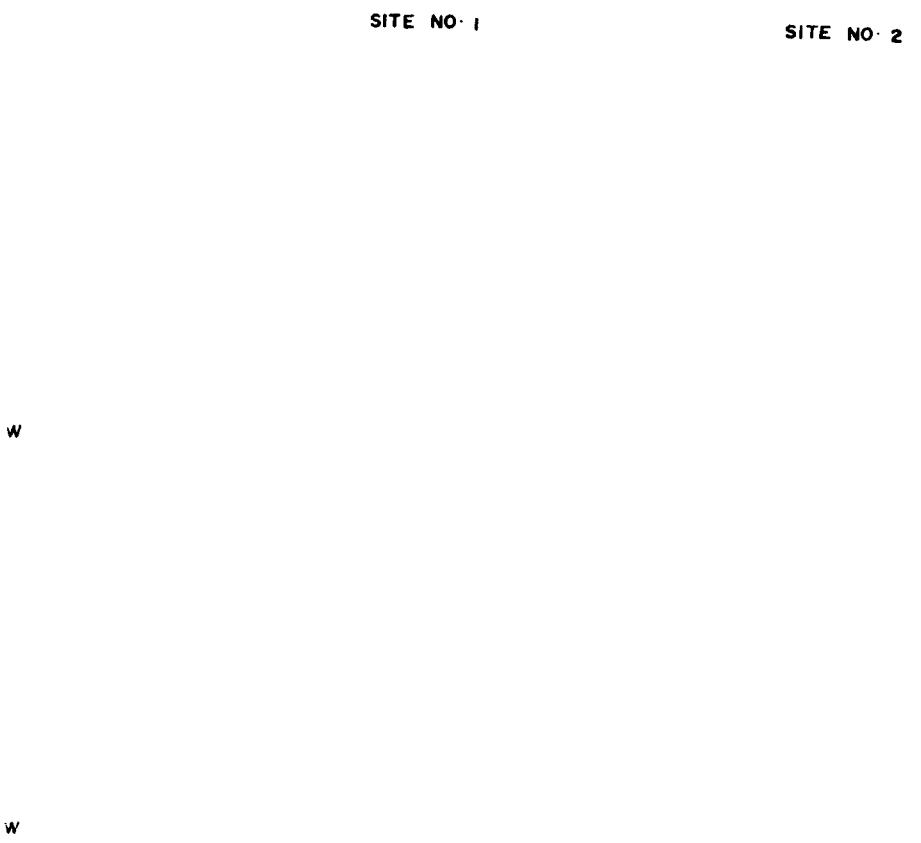



Fig. 7. Mean directions of the samples after thermal cleaning for each site plotted on equal area projection.

TABLE 3

Paleomagnetic parameters for the Kamthi sandstones, Rajmahal traps and Sylhet traps

| Locality | Site no. | $N$ | $D$ (°E) | $I$ (°) | $K$ | $R$ | $\Delta$ | VGP (North pole) $\lambda_p$ (°N) | $L_p$ (°W) | $d_p$ | $d_m$ | $\lambda$ (°S) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Kamthis 18°25′N 79°35′E | 1 | 12 | 303 | −44 | 65 | 11.83 | — | 18 | 48 | 4° | 6.2° | 26 |
| 18°82′N 79°35′E | 2 | 23 | 306 | −53 | 220 | 22.9 | — | 15 | 56 | $2.2_p$ | 2.8° | 34 |
| 19°7′N 79°30′E | 3 | 17 | 309 | −47 | 160 | 16.9 | — | 22 | 52 | 2.5° | 4° | 28 |
| Mean (Kamthis) | | — | 306 | −48 | 200 | — | 5.7° | 18 | 53 | 4.8° | 7.5° | 31 |
| Rajmahal traps | 15 | 92 | 322 | −64 | 170 | — | 6.3° | 12 | 66 | 3.5° | 4.3° | 46 |
| Sylhet traps | | 25 | 332 | −59 | 7 | 23.5 | — | 16 | 60 | 8° | 11° | 40 |

Symbols as in table 2.

Fig. 8. Virtual geomagnetic pole (VGP) positions for Parsora ($P_1$, $P_2$, $P_3$) and Kamthi formations ($K_1$, $K_2$, $K_3$), Rajmahal (R) and Sylhet (S) traps (data given in table 3).

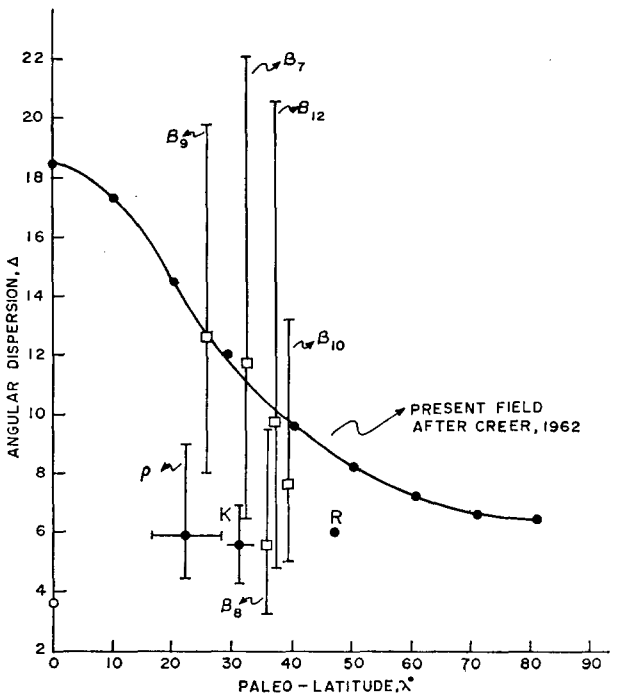



Fig. 9. Angular dispersion factor versus paleolatitude for the Parsora formation P (Upper Triassic age), the Kamthi formation K (Upper Permian age) and the Rajmahal traps (R) of Jurassic Estimates of angular dispersion factor for African formations as reported by McElhinny et al. (1968) are also given. $\beta_7$, $\beta_9$, etc. refer to poles reported by McElhinny et al. (1968).

For the calculation of the mean pole position, unit weight was given to each site. The virtual geomagnetic pole positions obtained for the three sites of Parsora are quite close to those obtained for Kamthis of Upper Permian age from Godavary Valley (Verma and Bhalla (1967). The pole position obtained so far for the Jurassic, Triassic and Permian formations in India is plotted in fig. 8 along with ovals of confidence at 95% probability level. Paleomagnetic parameters of interest for Rajmahal and Sylhet traps as well as Kamthis are listed in table 3.

Some estimate of the paleosecular variation can be obtained from the angular dispersion factor $\Delta = 81/\sqrt{k}$ where "$k$" is an estimate of Fisher's (1953) precision for these directions. The angular dispersion varies with latitude, and its magnitude for the present best fitting dipole field has been estimated by Creer (1952) as shown in fig. 9. For the rock formations which have been studied thoroughly it is of interest to obtain estimates of paleosecular variation from estimates of the precision of directions. A comparison of the present and the past estimates of the secular variation helps us to know whether the Earth's magnetic field has been always that of a geocentric dipole. The Parsora formation (Upper Triassic age), Kamthis (Upper Permian age) and Rajmahal traps (Jurassic age) have been sampled adequately so as to give an estimate of paleosecular variation. The $\Delta$ values for these formations are plotted in fig. 9 versus paleolatitude of sites. The 95% confidence limit of $\Delta$ was calculated according to a method given by Gough *et al.* (1964).

The estimates of paleosecular variation for African Mesozoic formations according to McElhinny *et al.* (1968) are also given in the figure for comparison. It appears that the magnitude of secular variation in India during Mesozoic was probably lower than that in Africa.

The paleomagnetic poles for the Parsora formation are different from those of the Rajmahal and Sylhet traps of Jurassic age as well as the Kamthi sandstones of Upper Permian age. A comparison of the mean Mesozoic pole position for India with those of other Gondwana continents is of interest for purposes of reconstructing Gondwanaland during Mesozoic. This mean has been reported earlier by McElhinny (1968) as 16°S, 121°E with $a_{95} = 10°$. A revised mean pole position for the Mesozoic is 21°30'S, 119°E, with

$a_{95} = 8°$, taking into consideration the data for the Tirupati sandstones formation of Lower to Middle Cretaceous age (VERMA and PULLAIAH (1967), the Rajmahal and Sylhet traps of Jurassic age (ATHAVALE *et al.* (1963), the KAMTHI sandstones of Upper Permian age (VERMA and BHALLA (1968) as well as the Parsora formation of Upper Triassic age. During Mesozoic times the poles relative to Africa appear to have remained fixed. According to MCELHINNY (1968) the mean Mesozoic pole for Africa is 65°S, 80°E. For Australia a similar situation appears to have existed from Carboniferous to Cretaceous times. The mean pole for the period "Kaiman to Mesozoic" for Australia has been given by IRVING (1966) as 48°S, 142°E with a 95% polar error of 5°. For the reconstruction of Gondwanaland, paleomagnetic data from India have been used by CREER (1965) as well as MCELHINNY (1968) amongst others. The later alone appears to have made use of the data for Upper Permian (Kamthi formation) from India. A reconstruction of Gondwanaland according to MCELHINNY *et al.* (1968) is shown in fig. 10. The revised mean Mesozoic pole suggests that the relative position of the continent of India may lie further west of Australia by about 6°.

Fig. 10. Reconstruction of Gondwanaland during Mesozoic according to MCELHINNY *et al.* (1968).

## References


ATHAVALE, R. N., C. RADHAKRISHNAMURTY and P. W. SAHASRABUDHE (1963) Geophys. J. **7**, 304.

ATHAVALE, R. N., R. K. VERMA, M. S. BHALLA and G. PULLAIAH (1968) Drift of the Indian subcontinent since precambrian times, Proc. of Nato Advance Study Inst., Conference in Paleogeophysics, Academic Press (in press).

CLEGG, J. A., E. R. DEUTCSH and P. H. GRIFFITHS (1956) Phil. Mag. **1**, 419.

CLEGG, J. A., C. RADHAKRISHNAMURTHY and P. W. SAHASRABUDHE (1958) Nature **181**, 830.

CREER, K. M. (1959) Geophys. J. **2**, 261.

CREER, K. M. (1962) J. Geophys. Res. **67**, 3461.

CHAMALAUN, F. H. and K. M. CREER (1963) Nature **198**, 375

COTTER, G. D. P. (1918) A revised classification of the Gondwana system, Rec. Geol. Sur. India, **48**, part 1.

DOELL, R. R. and A. COX (1967) Paleomagnetic sampling with a portable coring drill, in: *Methods in paleomagnetism* (Elsevier Publ. Co., Amsterdam) 21.

DEUTSCH, E. R., C. RADHAKRISHNAMURTY and P. W. SAHASRABUDHE (1959) Ann. Geophys. Paris **15**, 38.

FISHER, R. A. (1953) Proc. Roy. Soc. London Ser. A **217**, 295.

GOUGH, D. I., N. D. OPDYKE and M. W. MCELHINNY (1964) J. Geophys. Res. **69**, 2509.

IRVING, E. (1963) J. Geophys. Res. **68**, 2283.

IRVING, E. (1966) J. Geophys. Res. **71**, 6025.

IRVING, E., W. A. ROBERTSON, P. M. STOTT, D. H. TARLING and M. A. WARD (1961) J. Geophys. Res. **66**, 1827.

IRVING, E. and N. D. OPDYKE (1965) Geophys. J. **9**, 153.

LIKHITE, S. D. and C. RADHAKRISHNAMURTHY (1965) Bull. NGRI **3**, 1.

MCELHINNY, M. W., J. C. BRIDEN, D. L. JONES and A. BROCK (1968) Rev. Geophys. **6**, 201.

OPDYKE, N. D. (1964) J. Geophys. Res. **69**, 2473.

SAHASRABUDHE, P. W. (1963) Paleomagnetism and the geology of the Deccan traps, Proc. Seminar Geophys. investigations in the Peninsular shield (Indian Geophys. Union, Hyderabad, India) 226.

SAHNI, M. R. and C. N. RAO (1958) A note on the correlation of the Parsora and Tiki beds of Vindhya Pradesh, Intern. Geol. Conf., Mexico, Commission De Gondwana, 95–103.

SCOTT, G. G. (1957) Rev. Sci. Instr. **28**, 270.

SEWARD, A. C. (1933) On some fossil plants from Parsora stage, Rec. Geol. Sur. India, **66**, part 2.

VERMA, R. K. and A. APPA RAO (1963) Paleomagnetism of decca traps of Vikarabad, Chincholi and Gulbarga areas, Proc. Seminar Geophys. Investigations in the Peninsular shield (Indian Geophys. Union, Hyderabad, India).

VERMA, R. K. and G. PULLAIAH (1967) Earth Planet. Sci. Letters **2** 310.

VERMA, R. K., G. PULLAIAH and M. S. BHALLA (1967) Paleomagnetism of Tirupati sandstones and Kamthi sandstone from Godavary Valley, Proc. Symp. Upper Mantle Project, Hyderabad, India.

VERMA, R. K. and M. S. BHALLA (1968) J. Geophys. Res. **73**, 703.